

AI POWERED FAKE NEWS DETECTION PROJECT

Fake news has become a major challenge in today's digital world. This project uses Machine Learning to classify news articles as Fake or Real. A web application was developed where users can enter news text and receive a prediction.

OBJECTIVES

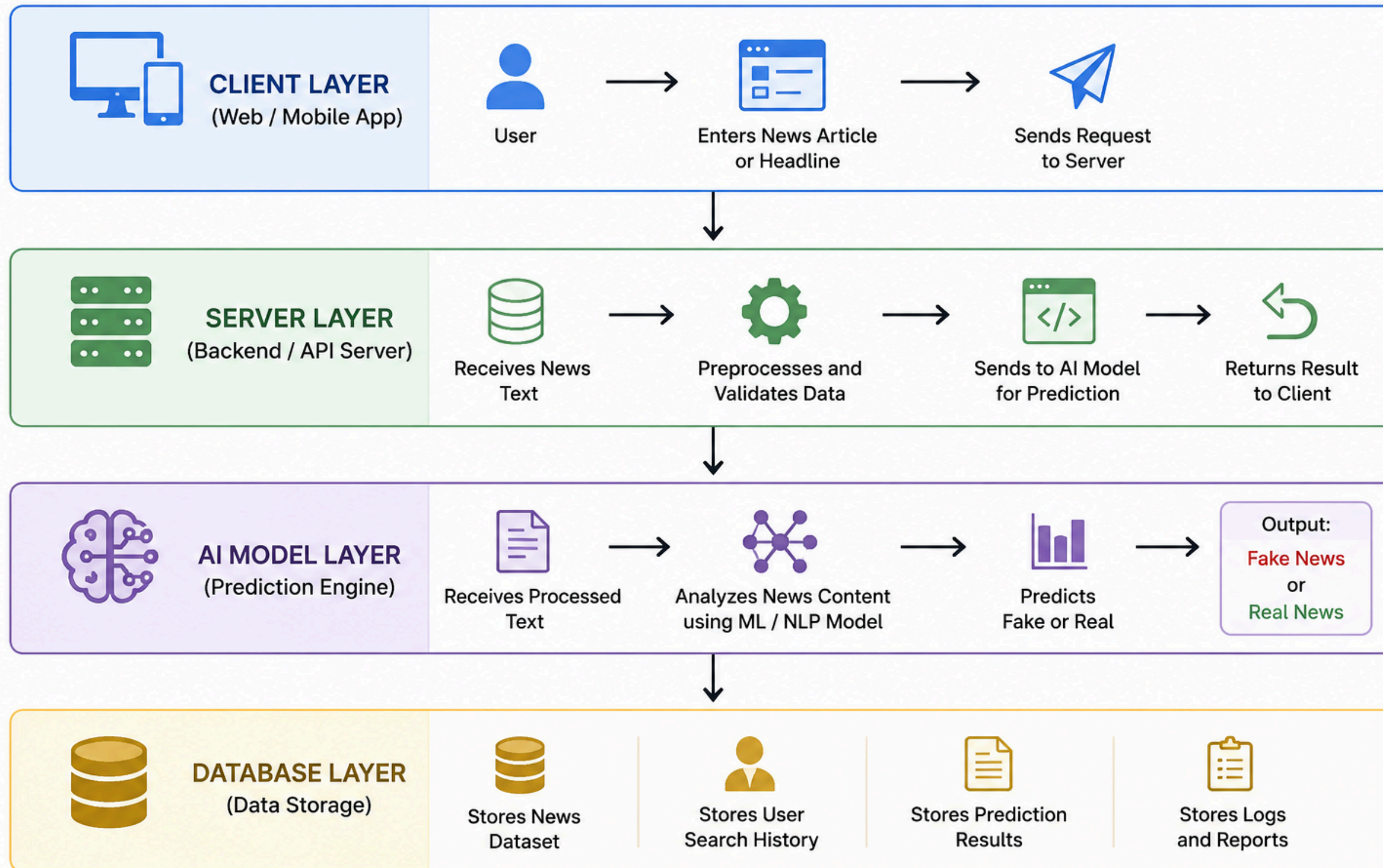
- DETECT FAKE NEWS USING MACHINE LEARNING.
- BUILD A USER-FRIENDLY WEB INTERFACE.
- PROVIDE FAST AND ACCURATE PREDICTIONS.

TECHNOLOGIES

- PYTHON -PROGRAMMING LANGUAGE
- FLASK - BACKEND
- HTML - FRONTEND
- CSS - STYLING
- JAVASCRIPT - USER INTERACTION
- SCIKIT-LEARN - MACHINE LEARNING
- PANDAS - DATA PROCESSING
- PICKLE - SAVE TRAINED MODEL

SYSTEM ARCHITECTURE

AI Fake News Detection – Client Server Architecture



MACHINE LEARNING PROCESS

1. LOAD DATASET
2. CLEAN TEXT
3. VECTORIZE TEXT USING TF-IDF
4. TRAIN MACHINE LEARNING MODEL
5. SAVE MODEL.PKL AND VECTORIZER.PKL
6. LOAD MODEL IN FLASK
7. PREDICT FAKE OR REAL

FUTURE IMPROVEMENTS

- IMPROVE PREDICTION ACCURACY.
- ADD SUPPORT FOR MULTIPLE LANGUAGES.
- DEPLOY THE APPLICATION ONLINE.
- USE DEEP LEARNING MODELS.

RESULTS

Web Application:- <https://truthlens-app.vercel.app/>

The screenshot displays the TruthLens web application in a browser. The browser's address bar shows the URL `truthlens-app.vercel.app`. The application's header includes the "TruthLens" logo and a "Backend online" status indicator. The main heading is "Is it real or fabricated?", followed by the instruction: "Paste any news article below. Our model analyses language patterns to detect misinformation."

The "ARTICLE INPUT" section contains a text area with the following text: "who is a national celebrity in his own right clarke gayford who hosts his own fishing television show has already broken the mold of his predecessors stepping in to defend his partner against scaremongering old dinos including one political rival who made a disparaging reference to arden and lipstick on a pig on twitter gayford has thrown fishing buddy jokes to winston peters the leader of the small nationalist party that decided arden's political fate and teased a nation that waited a month for an election result heading out of reception for a while things always seem to happen". A character count of "781 characters" is shown at the bottom right of the text area. Below the text area is a word count of "128 words" with a green progress bar. A blue information icon is followed by the text: "For best results, paste a complete news article or detailed news report. Short headlines or random sentences may not produce reliable predictions."

At the bottom of the input section are two buttons: "Clear" and "Analyse Article →".

The "VERDICT" section features a green checkmark icon, the text "REAL NEWS" in green, and the message: "The model found no strong indicators of misinformation in this article." Below this is a button labeled "Analyse another article".

RESULTS

▼

TruthLens — AI Fake News Dete ×

TruthLens — AI Fake News Dete ×

+

Ask Gemini

—

📄

×

←

→

🔄

🔍 truthlens-app.vercel.app

🔍

☆

👤

⋮

TruthLens

Backend online

AI-POWERED VERIFICATION

Is it real
or fabricated?

Paste any news article below. Our model analyses language patterns to detect misinformation.

ARTICLE INPUT

when hiring from the beginning or through march the department went on a hiring spree of deputies as part of its ongoing reformthe department will also have to go through old applications to find applicants who were eliminated because of their citizenship status and reconsider them for future jobsthe justice department made the announcement on monday saying the sheriff s department violated the immigration and nationality act without having an exemptionread more we

1,390 characters

210 words

For best results, paste a complete news article or detailed news report. Short headlines or random sentences may not produce reliable predictions.

Clear

Analyse Article →

VERDICT

×

FAKE NEWS

The model detected patterns commonly associated with fabricated or misleading content.

GITHUB REPOSITORY

GitHub Repository link :- <https://github.com/jiyapashte0-cloud/Al-Powered-Fake-News-Detection>

The screenshot displays the GitHub interface for the repository 'AI-Powered-Fake-News-Detection'. At the top, it shows the repository name, a 'Public' badge, and interaction buttons for 'Watch' (1), 'Fork' (0), and 'Star' (0). Below this, the 'main' branch is selected, with '1 Branch' and '0 Tags' indicated. A search bar and 'Add file'/'Code' buttons are present. The central file list shows recent commits by 'jiyapashte0-cloud', including updates to '.ipynb_checkpoints', 'backend', 'data', 'deployment', 'docs', 'frontend', 'reports', 'src', 'Fake_news_detection.ipynb', 'README.md', 'cleaned_dataset.csv', 'model.pkl', and 'vectorizer.pkl'. The right sidebar contains sections for 'About' (describing the project as 'AI-Powered Fake News Detection using python'), 'Releases' (no releases published), 'Packages' (no packages published), 'Contributors' (listing 'jiyapashte0-cloud' and 'MrunTheDev'), and 'Languages' (showing a bar chart for Jupyter Notebook (68.6%), CSS (11.4%), Python (8.7%), JavaScript (8.0%), and HTML (3.3%)). The bottom section shows the 'README' file content, which includes the title 'AI-Powered-Fake-News-Detection' and the subtitle 'AI-Powered Fake News Detection using python'.

AI-Powered-Fake-News-Detection

AI-Powered Fake News Detection using python

CONCLUSION

THIS PROJECT SUCCESSFULLY DEVELOPED AN AI-POWERED FAKE NEWS DETECTION SYSTEM USING MACHINE LEARNING TECHNIQUES. THE APPLICATION IS CAPABLE OF ANALYZING NEWS TEXT AND CLASSIFYING IT AS FAKE OR REAL THROUGH A USER-FRIENDLY WEB INTERFACE. THE PROJECT INVOLVED DATA PREPROCESSING, MODEL TRAINING, BACKEND DEVELOPMENT WITH FLASK, AND FRONTEND INTEGRATION USING HTML, CSS, AND JAVASCRIPT.

THE DEVELOPED SYSTEM DEMONSTRATES HOW ARTIFICIAL INTELLIGENCE CAN BE APPLIED TO ADDRESS THE GROWING CHALLENGE OF MISINFORMATION BY PROVIDING QUICK AND RELIABLE PREDICTIONS. THROUGHOUT THE PROJECT, PRACTICAL KNOWLEDGE OF MACHINE LEARNING, WEB DEVELOPMENT, API INTEGRATION, AND COLLABORATIVE SOFTWARE DEVELOPMENT USING GITHUB WAS GAINED.

ALTHOUGH THE CURRENT SYSTEM PERFORMS EFFECTIVELY, IT CAN BE FURTHER ENHANCED BY IMPROVING MODEL ACCURACY, SUPPORTING MULTIPLE LANGUAGES, INTEGRATING REAL-TIME FACT-CHECKING SERVICES, AND DEPLOYING THE APPLICATION ON A CLOUD PLATFORM FOR PUBLIC ACCESS. OVERALL, THIS PROJECT PROVIDES A STRONG FOUNDATION FOR DEVELOPING INTELLIGENT SYSTEMS THAT HELP USERS IDENTIFY POTENTIALLY FALSE INFORMATION AND PROMOTE ACCESS TO TRUSTWORTHY NEWS.

THANK YOU

PRESENTED BY (GROUP 21)

MRUNMAYI SURVE

JIYA PASHTE

ADITI DHUMAL

RUPALI RATHOD

TANVI MARKAD

GRISHMA INGALE